

Who You Are and Who You Learn From: Effects of Self-Similarity on Embodiment and Language Learning

Milan Wolff*
University of Wyoming

Jennifer LaVanchy†
University of Wyoming

Amy Banić‡
University of Wyoming



Figure 1: Conditions for the study, with the avatar shown in the mirror and the instructor standing behind the table. From left to right: SA/SI, DA/SI, SA/DI, DA/DI

ABSTRACT

The use of self-similar virtual humans in foreign language learning an underexplored area in VR research. In this 2x2 within-subjects study, we investigate the interacting effects of self-similarity in virtual instructor and user avatar design to promote short-term vocabulary recall in Spanish. Participants ($n = 38$) completed four introductory level Spanish language lessons. Self-similar avatars promoted significantly greater user-avatar identification and embodiment, in addition to higher subjective sense of similarity. Meanwhile, self-dissimilar instructors resulted in greater improvement in scores between pre-and post-tests. Our findings illuminate guidelines for designing avatars and instructors for language learning in VR.

Index Terms: Virtual reality, avatars, virtual humans, embodiment, language learning

1 INTRODUCTION

Virtual reality (VR) offers immersive experiences beyond those encountered in everyday life, including immersion in languages to individuals who may not have fluent speakers of the target language in their community or the resources to travel to different regions. Based on numerous studies in the language learning field, immersion methods yield better outcomes for people acquiring new languages [3], and empirical evidence demonstrates the positive impacts of immersive VR on foreign language learning [6]. However, the best practices for designing virtual reality experiences for language learning have yet to be established [5].

The appearance of a user's avatar, or the "digital representation of the self," particularly in "3D virtual environments," [8] is a factor not yet thoroughly understood within the context of language learning in virtual reality. In this study, we examine both the effects of self-similar, defined here as having physical characteristics visually resembling the user [1], and self-dissimilar, appearances of both avatar and instructor on two outcomes: user-avatar identification, which describes user connection with and embodiment as an avatar [2], and short-term recall of introductory-level Spanish vocabulary, as short-term vocabulary recall correlates with overall vocabulary retention in the target language [9].

*e-mail: mwolff3@uwyo.edu

†e-mail: lavanchy@uwyo.edu

‡e-mail: abanic@uwyo.edu

	Instructor (self-similar)	(self-dissimilar)
Avatar (self-similar)	SA/SI	SA/DI
(self-dissimilar)	DA/SI	DA/DI

Table 1: Combinations of avatar and instructor conditions used in this experiment

We investigate the following research questions about the use of self-similar virtual humans and avatars for language learning:

- Do self-similar avatars promote greater user-avatar identification?
- Do self-similar virtual humans and/or avatars improve important language learning outcomes such as short-term vocabulary recall?

2 EXPERIMENTAL DESIGN

We designed a within-subjects experiment with a 2x2 design outlined in the table below, with conditions presented in order according to a balanced Latin square design. We administered a pre- and post-test on the vocabulary terms presented in each lesson. Lesson vocabulary topics (fruits, colors, numbers, shapes) were randomized between conditions. Participants completed a questionnaire on user-avatar identification after each lesson [2]. Following examples from other studies on embodiment [4] [7], a virtual mirror was positioned in front of the user, allowing them to see their avatar during the experiment to increase levels of embodiment.

2.1 Participant Demographics

We collected data from $n = 38$ participants (21 men, 15 women, 2 non-binary people) ranging in age from 18 to 56 (28.32 ± 8.40 years). Twenty-one participants reported using VR ten or fewer times. All participants reported knowing English, and 21 reported some level of prior experience with Spanish.

2.2 Data Analysis

We performed repeated measures ANOVAs, which are appropriate for multiple measures collected from the same individuals, to assess the impact of different conditions on user-avatar identification and short-term vocabulary recall outcomes. We examined ordering effects for all variables and did not find any impact of ordering on the results presented, as we implemented a balanced Latin square design and randomization of lesson content to control for ordering effects in the experimental design.

2.3 Power Analysis

We conducted an a priori power analysis using the G*Power software package. We specified an effect size of 0.8 and α of 0.05,

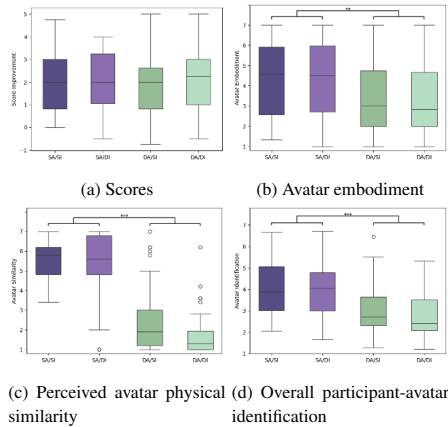


Figure 2: Results by avatar/instructor combination for scores and across different categories of user-avatar identification yielding a required sample size of $n = 18$, which was far exceeded in this study.

2.4 Short-Term Vocabulary Recall Outcomes

We used the improvement in number of correct vocabulary terms between pre- and post-tests to assess short-term vocabulary recall. The average score improvement between pre-test and post-test was 2.02 ± 1.25 for the SA/SI condition, 1.77 ± 1.42 for the DA/SI condition, 2.00 ± 1.29 for the SA/DI condition, 1.97 ± 1.35 for the DA/DI condition. We obtained ($F(3, 111) = 2.669$, $p = 0.05$, $\eta^2 = 0.067$) following normality corrections, indicating a moderate effect size and results near the threshold of significance (Figure ??).

2.5 Avatar Identification

We found a statistically significant difference in user-avatar identification between at least two groups ($F(3, 111) = 5.44$, $p = 0.003$, $\eta^2 = 0.068$) after correcting for normality violations.

2.5.1 Physical Resemblance

We found a statistically significant difference in user perceptions of physical resemblance to the avatar between at least two groups ($F(3, 111) = 46.93$, $p < 0.001$, $\eta^2 = 0.559$) after applying sphericity and normality corrections, indicating an extremely large effect size.

2.5.2 Embodiment

We found a statistically significant difference in embodiment of the avatar between at least two groups ($F(3, 111) = 7.71$, $p < 0.001$, $\eta^2 = 0.172$) after applying normality corrections, indicating a large effect size.

3 DISCUSSION

Based on the results above, self-similar avatars promote greater user-avatar identification. These differences appear across the embodiment and perceived physical self-similarity subscales of the polythetic user-avatar identification scale. Furthermore, there is some evidence to suggest that differences exist between combinations of self-similar and self-dissimilar avatars and instructors. Notably, conditions with a self-similar avatar showed greater score improvements than conditions with self-dissimilar avatars, and conditions with self-dissimilar virtual human instructors showed greater score improvements than conditions with self-similar virtual instructors. All participants learned additional Spanish vocabulary terms as a result of participation, further solidifying conclusions from previous research that virtual reality language learning environments can be effective for teaching languages. Short-term Spanish vocabulary recall was enhanced in the conditions with a self-similar avatars relative to conditions with self-dissimilar avatars and

in conditions with self-dissimilar instructors in comparison to self-similar instructors.

Our findings confirm previous research on the impacts of self-similarity on embodiment [1] and suggest that these findings generalize to other aspects of user identification with an avatar.

4 CONCLUSIONS

The interacting effects of virtual instructor appearance and avatar appearance in the context of language learning outcomes have not been studied extensively. We report greater user-avatar identification across different subscale and confirm the beneficial effects of avatar self-similarity on embodiment in this context. In addition, we verify that self-similar avatar designs are indeed perceived as having high physical resemblance to the user. We demonstrate that the use of self-dissimilar instructors, especially in conjunction with self-similar avatars, may increase introductory-level short-term Spanish vocabulary recall. Consequently, we recommend the use of self-similar avatars and self-dissimilar instructors to promote language learning. Future research directions include use of such avatars and virtual instructors in longitudinal studies, with other languages, or in other learning domains.

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